Project Report: Optimization with a Genetic Algorithm

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1 Introduction

1.1 Problem Definition

The objective of this project is to implement and analyze the performance of a Genetic Algorithm (GA) designed to solve a continuous optimization problem. Specifically, the task requires finding the global minimum of a complex two-dimensional mathematical function within a predefined search space (Area of Interest).

1.2 Genetic Algorithms (GA)

Genetic Algorithms are an artificial intelligence technique belonging to the field of evolutionary computation, inspired by Charles Darwin's principles of natural selection and genetics. The algorithm operates on a *population* of individuals, each representing a potential solution to the problem. Each individual possesses a *chromosome* (in our case, a 32-bit binary string). Through iterative processes of *selection* (based on fitness), *crossover* (recombination of genetic material), and *mutation* (random alteration of bits), the population evolves generation after generation, progressively converging towards optimal solutions.

2 The Objective Function

The function for which the global minimum is to be found is defined as follows:

$$f(x_1, x_2) = 1.5 - \exp(-x_1^2 - x_2^2) - 0.5\exp(-(x_1 - 1)^2 - (x_2 + 2)^2) \tag{1}$$

This function is characterized by a complex topography that includes local minima where a classic optimization algorithm might get trapped. The Area of Interest (AOI) set for the search is defined in the continuous interval $[-5, 5]$ for both variables x_1 and x_2. The genetic algorithm is responsible for decoding the 32-bit chromosome (16 bits per variable) to map it exactly within this continuous space. Mathematical analysis of the objective function reveals two key basins of attraction: a global minimum at $f(0, 0) \approx 0.4966$ and a shallower local minimum (a "trap") at $f(1, -2) \approx 0.9933$. Establishing these theoretical values is crucial: they serve as our *ground truth*. If our Genetic Algorithm converges near 0.4966, it proves it successfully navigated the complex topography to find the true optimum, whereas converging near 0.9933 would indicate it was caught in the local trap.

3 Parameter Configuration and Design Choices

The algorithm's configuration balances computational efficiency with solution accuracy. During testing, unmodified parameters were kept at their base values.

3.1 Fixed Parameters (Structural)

- **Search Domain (AOI) at $[-5, 5]$:** Force the algorithm to start in peripheral, flat-gradient areas proves its robustness in finding the correct basins of attraction without relying on a priori assumptions.
- **Chromosome Length (32 bits):** Allocating 16 bits per variable over a search interval of 10 provides a high spatial precision ($\Delta x \approx 0.00015$).
- **Number of Generations (50):** Set as a computational budget. Tests showed that the fitness curve stabilizes well before this limit, avoiding redundant calculations on an already converged population.

3.2 Dynamic Parameters (Optimal Values)

Based on the Parameter Study (detailed in Section 4), the following base values optimally balance exploration and exploitation:

- **Population (100) & Tournament Size (3):** A population of 100 ensures sufficient genetic diversity. A tournament size of 3 applies a moderate selective pressure, preventing premature convergence by keeping useful traits from sub-optimal solutions alive.
- **Crossover (80%) & Mutation (5%):** A high crossover rate efficiently exploits promising combinations. Concurrently, a 5% mutation rate injects just enough entropy to escape local minima without degrading the process into a pure random search.

4 Results and Plot Analysis (Parameter Study)

For each tested configuration, the algorithm was executed 5 times to calculate the mean of the found minima and the standard deviation, thus neutralizing the effect of initial randomness.

4.1 Mutation Probability

The mutation probability regulates the exploration of the search space by reintroducing genetic diversity.

Value	Mean Minimum	Standard Deviation
0.01	0.49800	0.00266
0.05	0.49661	0.00002
0.20	0.49659	0.00002

Table 1: Mutation Probability Test Results

Analysis: With a very low mutation rate (1%), a slightly worse mean (0.49800) and a higher standard deviation are noted. The algorithm does not get trapped in the local minimum (0.9933), but it suffers from *premature convergence* within the global optimum's basin. Because the mutation rate is too low, the population exhausts its genetic diversity and becomes a mass of clones before reaching the exact mathematical bottom (0.4966), losing the ability to fine-tune the solution. The values of 5% and 20% provide the necessary genetic variations to reach the exact optimum, but 5% proves to be the ideal compromise to ensure stability without devolving into pure random exploration.

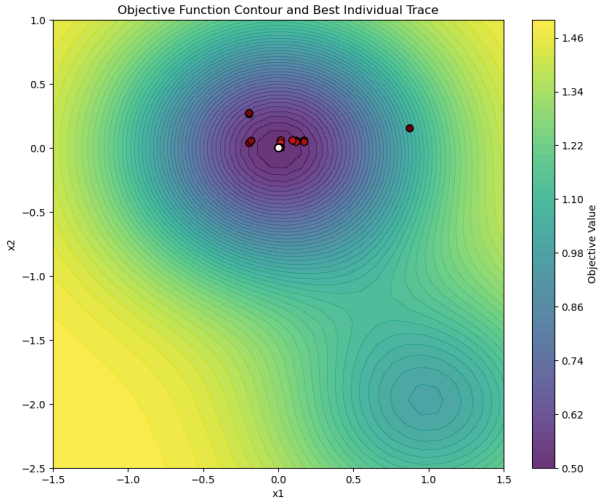


Figure 1: Best individual trace with 5% Mutation

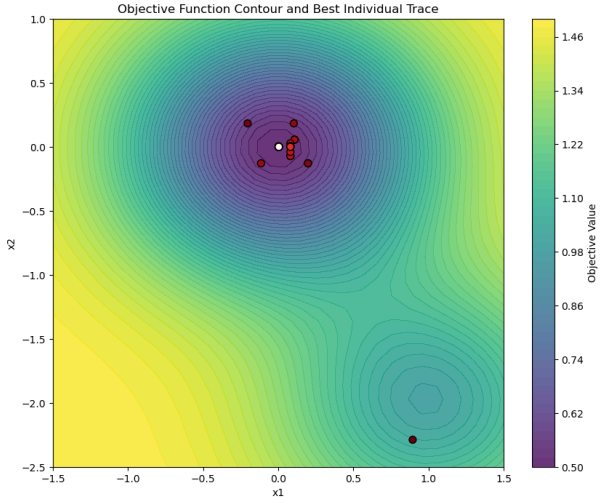


Figure 2: Best individual trace with 1% Mutation

4.2 Tournament Size

Tournament size regulates selective pressure: how severe the selection of parents for reproduction is.

Value	Mean Minimum	Standard Deviation
2	0.49661	0.00002
5	0.49659	0.00002
10	0.49661	0.00001

Table 2: Tournament Size Test Results

Analysis: The results indicate that the algorithm is exceptionally robust to variations in selective pressure for this specific topology. A tournament size of 2 provides a gentler selection (preserving diversity), while a size of 10 applies aggressive selective pressure (favoring rapid convergence). Despite these differences, all three configurations successfully locate the exact global minimum (0.4966). The slightly lower standard deviation at $TournamentSize = 10$ (0.00001) suggests that higher selective pressure accelerates the culling of sub-optimal individuals without triggering premature convergence. This resilience is largely due to the base mutation rate (5%) and population size (100), which provide a sufficient counterbalance to maintain genetic diversity even under severe selection.

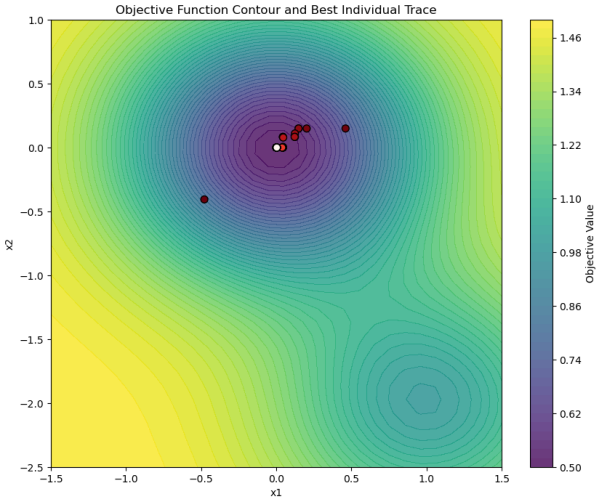


Figure 3: Best individual trace with Tournament = 5

4.3 Population Size

Determines the amount of genetic material available at each generation.

Value	Mean Minimum	Standard Deviation
20	0.69015	0.23449
100	0.49661	0.00002
200	0.49659	0.00002

Table 3: Population Size Test Results

Analysis: This test highlighted the most critical issues. A reduced population of 20 individuals consistently fails (Mean 0.690, very high Deviation 0.234); the scarcity of initial DNA prevents the algorithm from adequately mapping the function. Increasing the population from 100 to 200 does not bring significant improvements to accuracy, but it doubles the computational costs. Therefore, a population of 100 is confirmed as the optimal balance.

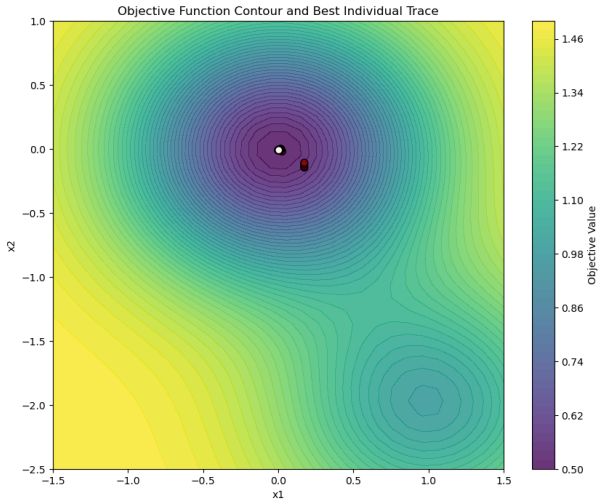


Figure 4: Best individual trace with Population = 100

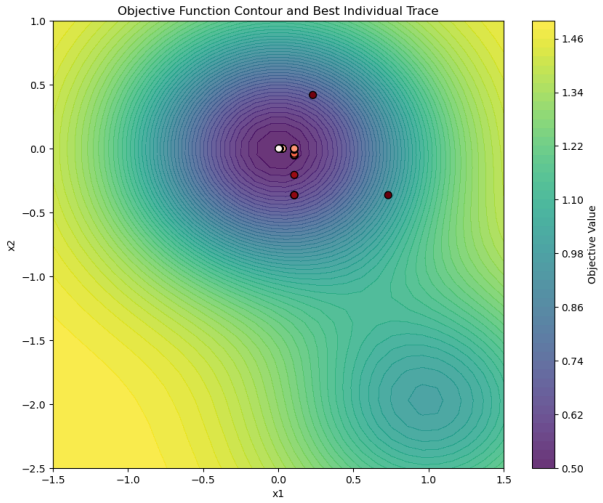


Figure 5: Best individual trace with Population = 20

4.4 Crossover Probability

Represents the probability that two parents will mix their genes to create new offspring.

Value	Mean Minimum	Standard Deviation
0.4	0.49660	0.00002
0.8	0.49660	0.00003
1.0	0.49661	0.00002

Table 4: Crossover Probability Test Results

Analysis: Crossover proves to be highly effective even at lower percentages for this specific problem. However, the standard value of 0.8 (80%) is recommended in the literature as it ensures excellent exploitation of partial solutions, while still allowing clones of highly performing individuals to pass to the next generation in case no mating occurs.

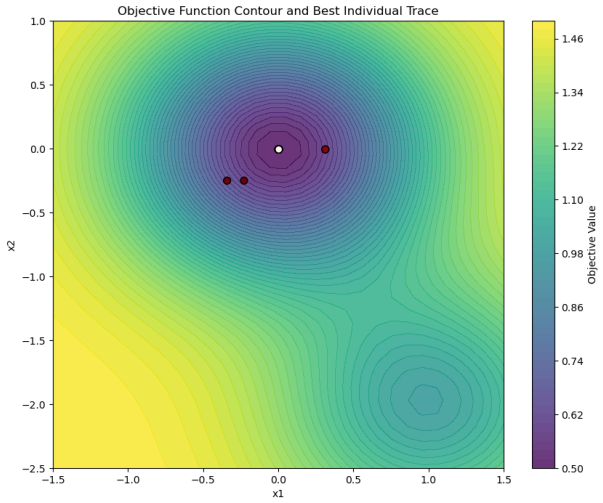


Figure 6: Best individual trace with 80% Crossover

5 General Conclusions

The implemented Genetic Algorithm demonstrated excellent resolution capabilities and remarkable stochastic stability, successfully navigating a complex topology to locate the exact global minimum. Beyond the numerical results, this project provided a valuable opportunity to deepen our understanding of evolutionary computation, effectively bridging the gap between the theoretical concepts explored during lectures and their practical application. Designing the algorithm and systematically tuning its hyperparameters proved to be a complex yet highly stimulating challenge. We experienced firsthand how the success of a GA relies heavily on the delicate balance between exploration and exploitation. The parameter study highlighted that while the algorithm is robust to many variations, a drastic reduction in population size fatally compromises the biodiversity necessary for spatial exploration. Ultimately, finding the correct base configuration not only ensured fast and reliable convergence, but also consolidated our engineering approach to solving complex optimization problems.